Quiz 1 Review

COSI 115b — Fundamentals of NLP II

February 27, 2026

Quiz 1 — Logistics

- Full class period
- 6 sections, 100 points total
- Answer in your blue book
- Show your work for all computation questions
- No calculators — leave answers as fractions where appropriate

Question types:

- True / False
- Multiple choice
- Short answer (conceptual)
- Computation / trace-through

Quiz 1 — Topics at a Glance

1. Parsing (CKY)
2. Perceptron
3. Logistic Regression
4. Distributional Semantics
5. Neural Networks
6. Structured Prediction

1. Parsing (CKY) — What to Know

Concepts:

- What is Chomsky Normal Form and why CKY requires it
- Time complexity of CKY and where it comes from
- How probabilistic CKY computes tree probabilities
- Ambiguity — can a sentence have multiple parses?

Skills:

- Given a PCFG in CNF and a sentence, fill in the full CKY chart from scratch
- Identify the most likely parse and its probability
- Draw the corresponding parse tree

1. Parsing (CKY) — Practice Setup

Grammar (CNF):

S → NP VP	[1.0]	Det → *the*	[1.0]
NP → Det N	[0.6]	N → *dog*	[0.5]
NP → N N	[0.4]	N → *park*	[0.5]
VP → V NP	[1.0]	V → *saw*	[1.0]

Sentence: *the dog saw the park*

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Sentence: *the dog saw the park*

Fill in a CKY chart. What is the most likely parse and its probability?

2. Perceptron — What to Know

Concepts:

- The perceptron convergence theorem — what it guarantees and when
- What linear separability means and why it matters
- The difference between standard and averaged perceptron
- Why averaging helps generalization

Skills:

- Given data and initial weights, trace through perceptron updates by hand
- Compute $\mathbf{w} \cdot \mathbf{x} + b$, apply sign, check correctness, update if wrong
- Know the update rule: $\mathbf{w} \leftarrow \mathbf{w} + y\mathbf{x}, \quad b \leftarrow b + y$

2. Perceptron — Practice Setup

Data:

$\mathbf{x}$	y
$(1, 2)$	$+1$
$(2, 0)$	-1
$(0, 1)$	$+1$

Initial weights: $\mathbf{w} = (0, 0),\ b = 0$

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$(2, 0)$	-1
$(0, 1)$	$+1$

Initial weights: $\mathbf{w} = (0, 0)$, $b = 0$

Trace one full pass through the data.

2. Perceptron — Worked Example

Start: $\mathbf{w} = (0, 0),\ b = 0$

Example 1: $\mathbf{x} = (1, 2),\ y = +1$

- $\mathbf{w} \cdot \mathbf{x} + b = 0.$ $\quad \text{sign}(0) = +1 = +1 \Rightarrow$ **no update**

2. Perceptron — Worked Example

Start: $\mathbf{w} = (0, 0),\ b = 0$

Example 1: $\mathbf{x} = (1, 2),\ y = +1$

- $\mathbf{w} \cdot \mathbf{x} + b = 0. \quad \text{sign}(0) = +1 = +1 \Rightarrow$ **no update**

Example 2: $\mathbf{x} = (2, 0),\ y = -1$

- $\mathbf{w} \cdot \mathbf{x} + b = 0. \quad \text{sign}(0) = +1 \neq -1 \Rightarrow$ **update**
- $\mathbf{w} = (0, 0) + (-1)(2, 0) = (-2, 0), \quad b = 0 + (-1) = -1$

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- $\mathbf{w} = (0, 0) + (-1)(2, 0) = (-2, 0), \quad b = 0 + (-1) = -1$

Example 3: $\mathbf{x} = (0, 1),\ y = +1$

- $\mathbf{w} \cdot \mathbf{x} + b = -2(0) + 0(1) + (-1) = -1. \quad \text{sign}(-1) = -1 \neq +1 \Rightarrow$ **update**
- $\mathbf{w} = (-2, 0) + 1(0, 1) = (-2, 1), \quad b = -1 + 1 = 0$

End of pass: $\mathbf{w} = (-2, 1),\ b = 0$

3. Logistic Regression — What to Know

Concepts:

- How logistic regression differs from the perceptron
- The sigmoid function: $\sigma(z) = \frac{1}{1+e^{-z}}$ and its properties
- What the model outputs (probabilities) vs. perceptron (labels)
- The gradient of the negative log-likelihood loss
- The SGD weight update rule
- Can logistic regression handle multiclass? (Yes — softmax)

Skills:

- Compute $\mathbf{w} \cdot \mathbf{x} + b$, apply sigmoid, get $\hat{y}$
- Write the gradient formula and compute it for a given example
- Write the update rule and compute updated weights

3. Logistic Regression — Practice Setup

Given: $\mathbf{w} = (1, -1)$, $b = 0$, $\eta = 1$

Training example: $\mathbf{x} = (2, 1)$, $y = 1$

3. Logistic Regression — Practice Setup

Given: $\mathbf{w} = (1, -1)$, $b = 0$, $\eta = 1$

Training example: $\mathbf{x} = (2, 1)$, $y = 1$

Perform one SGD update step.

Useful values: $\sigma(1) \approx 0.73, \quad e \approx 2.72$

3. Logistic Regression — Worked Example

Step 1: Forward pass

$$z = \mathbf{w} \cdot \mathbf{x} + b = 1(2) + (-1)(1) + 0 = 1$$

$$\hat{y} = \sigma(1) \approx 0.73$$

3. Logistic Regression — Worked Example

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$$\hat{y} = \sigma(1) \approx 0.73$$

Step 2: Compute gradient

$$\nabla_{\mathbf{w}} \mathcal{L} = (\hat{y} - y)\mathbf{x} = (0.73 - 1)(2, 1) = (-0.54,\ -0.27)$$

$$\frac{\partial \mathcal{L}}{\partial b} = (\hat{y} - y) = -0.27$$

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Step 3: Update

$$\mathbf{w}' = \mathbf{w} - \eta \nabla_{\mathbf{w}} \mathcal{L} = (1, -1) - 1 \cdot (-0.54, -0.27) = (1.54,\ -0.73)$$

$$b' = 0 - 1 \cdot (-0.27) = 0.27$$

4. Distributional Semantics — What to Know

Concepts:

- The distributional hypothesis — state it precisely
- Co-occurrence matrices and what they represent
- Pointwise Mutual Information (PMI) — formula and intuition
- Why we use Positive PMI (PPMI) over raw PMI
- Cosine similarity — what it measures

Skills:

- Compute cosine similarity between word vectors by hand
- Write the PMI formula
- Interpret what cosine similarity values mean

4. Distributional Semantics — Practice Setup

Cosine similarity:

Given word vectors: $\mathbf{a} = (1, 3, 0)$ and $\mathbf{b} = (2, 1, 0)$

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Compute $\cos(\mathbf{a}, \mathbf{b})$.

PMI:

$P(\text{dog}) = 0.01, \quad P(\text{bark}) = 0.005, \quad P(\text{dog}, \text{bark}) = 0.002$

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Cosine similarity:

Given word vectors: $\mathbf{a} = (1, 3, 0)$ and $\mathbf{b} = (2, 1, 0)$

Compute $\cos(\mathbf{a}, \mathbf{b})$.

PMI:

$P(\text{dog}) = 0.01, \quad P(\text{bark}) = 0.005, \quad P(\text{dog}, \text{bark}) = 0.002$

Compute $\text{PMI}(\text{dog}, \text{bark})$.

4. Distributional Semantics — Worked Example

Cosine similarity:

$$\mathbf{a} \cdot \mathbf{b} = 1(2) + 3(1) + 0(0) = 5$$

$$\|\mathbf{a}\| = \sqrt{1 + 9 + 0} = \sqrt{10}, \quad \|\mathbf{b}\| = \sqrt{4 + 1 + 0} = \sqrt{5}$$

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{5}{\sqrt{10} \cdot \sqrt{5}} = \frac{5}{\sqrt{50}} = \frac{5}{5\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.707$$

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PMI:

$$\text{PMI}(\text{dog}, \text{bark}) = \log_2 \frac{P(\text{dog}, \text{bark})}{P(\text{dog}) \cdot P(\text{bark})} = \log_2 \frac{0.002}{0.01 \times 0.005} = \log_2 \frac{0.002}{0.00005} = \log_2 40 \approx 5.32$$

5. Neural Networks — What to Know

Concepts:

- Why non-linear activations are necessary
- What happens with only linear layers (collapses to single layer)
- ReLU: definition and behavior
- Backpropagation: direction of gradient flow
- More layers $\neq$ always better (overfitting, vanishing gradients)

Skills:

- Compute a forward pass through an MLP: $\mathbf{z} = W\mathbf{x} + \mathbf{b}$, apply activation, repeat
- Write the chain rule decomposition for $\frac{\partial \mathcal{L}}{\partial W^{(k)}}$ at any layer

5. Neural Networks — Practice Setup

Two-layer MLP with ReLU:

$$W^{(1)} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{b}^{(1)} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$\mathbf{w}^{(2)} = \begin{pmatrix} 1 & 1 \end{pmatrix}, \quad b^{(2)} = 0$$

Input: $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

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Input: $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Compute the forward pass.

5. Neural Networks — Worked Example

Layer 1:

$$\mathbf{z}^{(1)} = W^{(1)}\mathbf{x} + \mathbf{b}^{(1)} = \begin{pmatrix} 1(2) + (-1)(1) \\ 0(2) + 2(1) \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

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$$\mathbf{h}^{(1)} = \text{ReLU}\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

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Layer 2:

$$z^{(2)} = \mathbf{w}^{(2)}\mathbf{h}^{(1)} + b^{(2)} = 1(1) + 1(1) + 0 = 2$$

Output: $\hat{y} = 2$

5. Neural Networks — Worked Example

Layer 1:

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Layer 2:

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Output: $\hat{y} = 2$

Chain rule for $\frac{\partial \mathcal{L}}{\partial W^{(1)}}$: $\quad \frac{\partial \mathcal{L}}{\partial W^{(1)}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial \mathbf{h}^{(1)}} \cdot \frac{\partial \mathbf{h}^{(1)}}{\partial \mathbf{z}^{(1)}} \cdot \frac{\partial \mathbf{z}^{(1)}}{\partial W^{(1)}}$

6. Structured Prediction — What to Know

Concepts:

- Local classification vs. structured prediction — why local fails
- The argmax problem: why it's intractable and what makes it feasible
- The feature function $\Phi(\mathbf{x}, \mathbf{y})$ and what it represents
- The Viterbi algorithm and its complexity
- The structured perceptron update rule and when it fires

Skills:

- Compute $\mathbf{w} \cdot \Phi(\mathbf{x}, \mathbf{y})$ for gold and predicted structures
- Determine if an update occurs (structure mismatch, not score mismatch)
- Apply the update: $\mathbf{w} \leftarrow \mathbf{w} + \Phi(\mathbf{x}, \mathbf{y}^*) - \Phi(\mathbf{x}, \hat{\mathbf{y}})$

6. Structured Prediction — Practice Setup

POS tagging with structured perceptron.

Sentence: *the dog runs*

Tags: {D, N, V}

- Gold: $\mathbf{y}^* = [\text{D, N, V}]$
- Predicted: $\hat{\mathbf{y}} = [\text{D, N, N}]$

Feature function Φ counts (word, tag) and $(\text{tag}_{i-1}, \text{tag}_i)$ bigrams.

6. Structured Prediction — Practice Setup

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Sentence: *the dog runs*

Tags: {D, N, V}

- Gold: $\mathbf{y}^* = [\text{D, N, V}]$
- Predicted: $\hat{\mathbf{y}} = [\text{D, N, N}]$

Feature function Φ counts (word, tag) and $(\text{tag}_{i-1}, \text{tag}_i)$ bigrams.

Does an update occur? If so, compute $\Phi(\mathbf{x}, \mathbf{y}^*) - \Phi(\mathbf{x}, \hat{\mathbf{y}})$.

6. Structured Prediction — Worked Example

$\hat{\mathbf{y}} \neq \mathbf{y}^* \Rightarrow$ **update occurs**

6. Structured Prediction — Worked Example

$\hat{\mathbf{y}} \neq \mathbf{y}^* \Rightarrow$ **update occurs**

$\Phi(\mathbf{x}, \mathbf{y}^*)$: [D, N, V]

- (the, D): 1
- (dog, N): 1
- (runs, V): 1
- (D→N): 1
- (N→V): 1

$\Phi(\mathbf{x}, \hat{\mathbf{y}})$: [D, N, N]

- (the, D): 1
- (dog, N): 1
- (runs, N): 1
- (D→N): 1
- (N→N): 1

6. Structured Prediction — Worked Example

$\hat{\mathbf{y}} \neq \mathbf{y}^* \Rightarrow$ **update occurs**

$\Phi(\mathbf{x}, \mathbf{y}^*)$: [D, N, V]

- (the, D): 1
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$\Phi(\mathbf{x}, \hat{\mathbf{y}})$: [D, N, N]

- (the, D): 1
- (dog, N): 1
- (runs, N): 1
- (D→N): 1
- (N→N): 1

$\Phi(\mathbf{x}, \mathbf{y}^*) - \Phi(\mathbf{x}, \hat{\mathbf{y}})$:

- (runs, V): $+1$, (runs, N): -1
- (N→V): $+1$, (N→N): -1

All other features cancel out.

General Tips

- **Show your work** — partial credit is easier to give if we can see your reasoning
- **Don't panic on computation** — the numbers are designed to be manageable by hand
- **Fractions are fine** — don't waste time on decimal approximations
- **Label clearly** — in your blue book, label each section and part (e.g., "3.2(c)")
- **Manage your time** — don't spend too long on one section

Good luck!